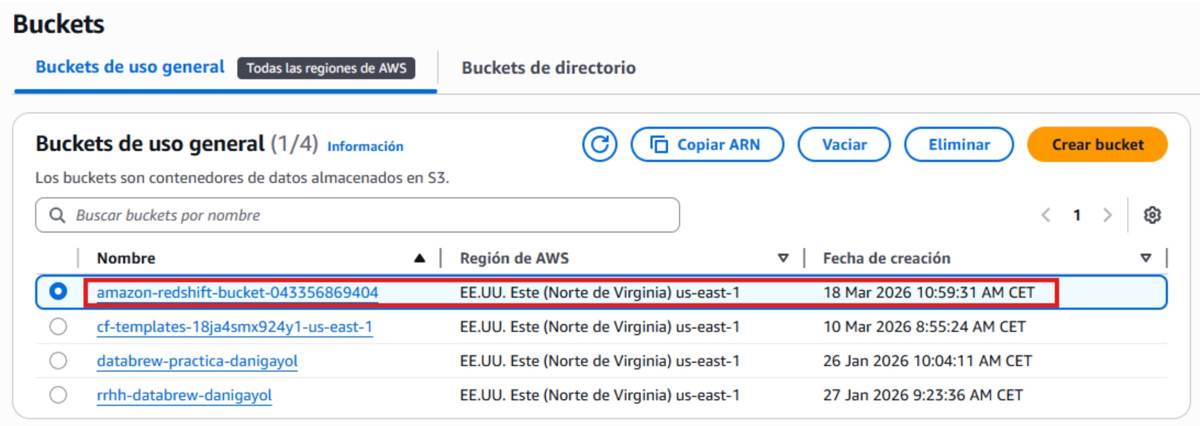
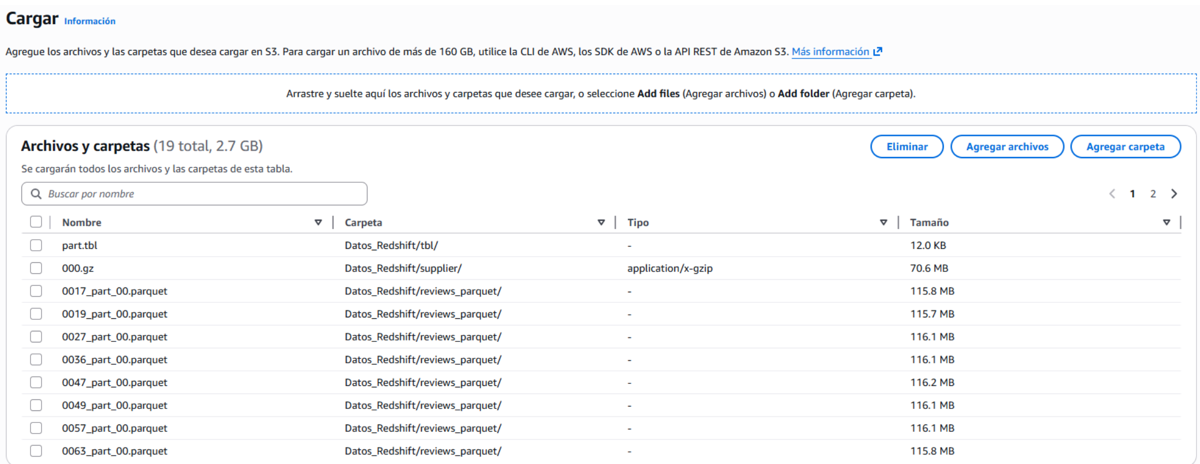
PR_11.4 Dani Gayol Rodríguez

Apartado A

Primeramente, tenemos que crear un Bucket en “S3”:

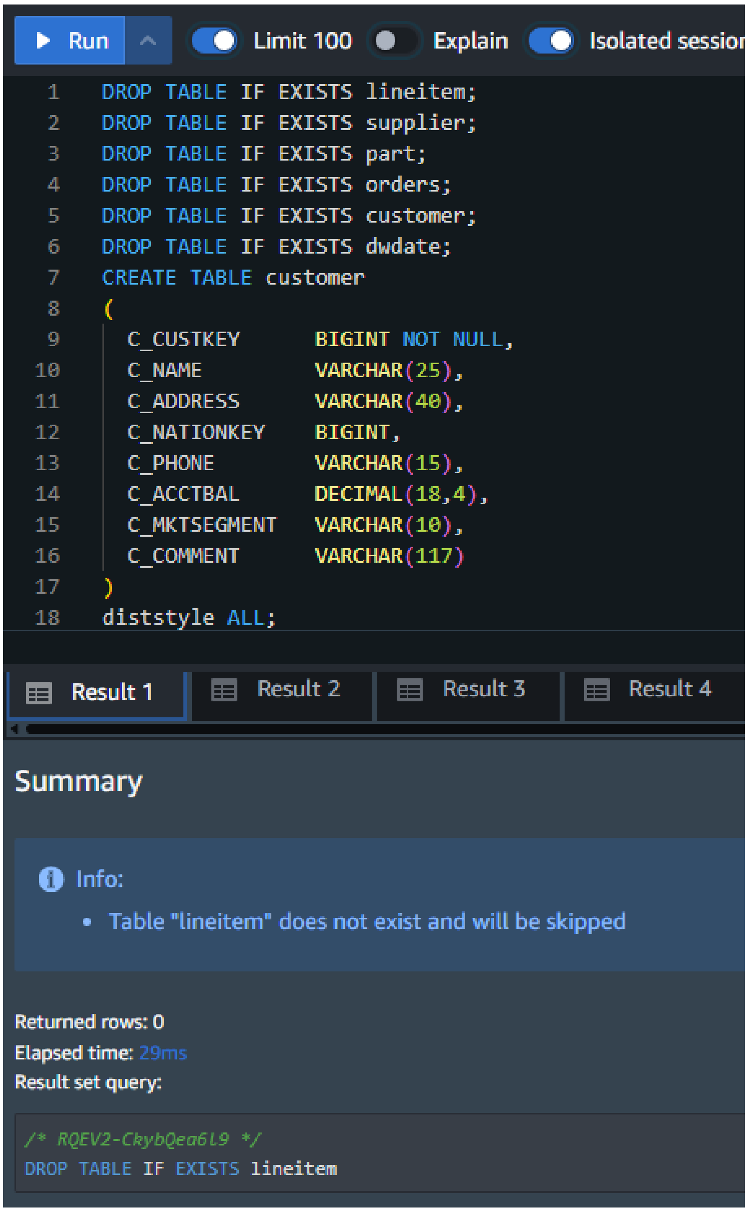


Ahora vamos a cargar los datos dentro del bucket:



Carga de datos desde Amazon S3 utilizando COPY

Ahora abrimos “Amazon Redshift” y ejecutamos el SQL:



Ahora vamos a hacer el "COPY":

```sql
COPY customer
from 's3://amazon-redshift-bucket-043356869404/Datos_Redshift/customer/'
iam_role 'arn:aws:iam::043356869404:role/LabRole'
CSV gzip;
COPY orders
from 's3://amazon-redshift-bucket-043356869404/Datos_Redshift/orders/'
iam_role 'arn:aws:iam::043356869404:role/LabRole'
CSV gzip;
COPY part
from 's3://amazon-redshift-bucket-043356869404/Datos_Redshift/part/'
iam_role 'arn:aws:iam::043356869404:role/LabRole'
CSV gzip;
COPY supplier
from 's3://amazon-redshift-bucket-043356869404/Datos_Redshift/supplier/'
iam_role 'arn:aws:iam::043356869404:role/LabRole'
CSV gzip;
COPY lineitem
from 's3://amazon-redshift-bucket-043356869404/Datos_Redshift/lineitem/'
iam_role 'arn:aws:iam::043356869404:role/LabRole'
CSV gzip;
COPY dwdate
from 's3://amazon-redshift-bucket-043356869404/Datos_Redshift/dwdate/'
iam_role 'arn:aws:iam::043356869404:role/LabRole'
CSV gzip dateformat 'auto';
```

Ahora comprobamos que se cargaron los datos correctamente:

```sql
select count(1) from lineitem;
select count(1) from supplier;
select count(1) from part;
select count(1) from orders;
select count(1) from customer;
select count(1) from dwdate;
```

Result 1 (1) | Result 2 (1) | Result 3 (1)

count
10006864

Ya podemos hacer las consultas:

```sql
SELECT c_mktsegment,
  COUNT(o_orderkey) AS orders_count,
  SUM(l_quantity) AS quantity,
  SUM(l_extendedprice) AS extendedprice,
  COUNT(DISTINCT P_PARTKEY) AS parts_count,
  COUNT(DISTINCT L_SUPPKEY) AS supplier_count,
  COUNT(DISTINCT o_custkey) AS customer_count
FROM lineitem
  JOIN orders ON l_orderkey = o_orderkey
  JOIN customer c ON o_custkey = c_custkey
  JOIN dwdate
  ON d_date = l_commitdate
  AND d_year = 1992
  JOIN part ON P_PARTKEY = l_PARTKEY
  JOIN supplier ON L_SUPPKEY = S_SUPPKEY
GROUP BY c_mktsegment;
```

Result 1 (5) Export

c mktsegment	orders count	quantity	extendedprice	parts count	supplier count	customer count
AUTOMOBILE	129	2962	4436256.23	118	118	111
MACHINERY	118	3344	4904436.41	107	107	99
HOUSEHOLD	117	3160	4807231.66	107	107	100
BUILDING	126	3477	5165523.39	117	117	110
FURNITURE	115	2995	4453895.46	105	105	96

```sql
CREATE TABLE product_reviews(
  marketplace varchar(2),
  customer_id varchar(32),
  review_id varchar(24),
  product_id varchar(24),
  product_parent varchar(32),
  product_title varchar(512),
  star_rating int,
  helpful_votes int,
  total_votes int,
  vine char(1),
  verified_purchase char(1),
  review_headline varchar(256),
  review_body varchar(max),
  review_date date,
  year int,
  product_category varchar(32)
)
DISTSTYLE KEY
DISTKEY (customer_id)
SORTKEY (
  marketplace,
  product_category,
  review_date);
```

Result 1

Summary

Returned rows: 0

Query ID: 613120

Elapsed time: 293ms

Result set query:

```sql
COPY product_reviews
FROM 's3://amazon-redshift-bucket-043356869404/Datos_Redshift/reviews_parquet/'
iam_role 'arn:aws:iam::043356869404:role/LabRole'
PARQUET;
```

Result 1

Summary

Info:

- Load into table 'product_reviews' completed, 5906460 record(s) loaded successfully.

Returned rows: 0

Elapsed time: 1m 22.5s

Result set query:

```sql
/* RQEV2-RywGhGhDJk */
COPY product_reviews
FROM 's3://amazon-redshift-bucket-043356869404/Datos_Redshift/reviews_parquet/'
iam_role 'arn:aws:iam::043356869404:role/LabRole'
PARQUET
```

```sql
SELECT product_title,
 SUM(total_votes)
FROM product_reviews
WHERE product_category = 'Apparel'
GROUP BY product_title
ORDER BY SUM(total_votes) DESC LIMIT 10;
```

Result 1 (10)

product_title	sum
THREE WOLF MOON S...	61869
Delicious PhD Darling Co...	59309
The Mountain Kids 100% ...	42144
The Mountain Three Wolf ...	26107
Squeem 'Perfect Waist' C...	14665
Ann Chery Women's Faja...	9536
Ann Chery Women's Faja...	8497
F500 American Flag Pant...	7378
MUXXN Women's 1950s ...	6728
Levi's Men's 501 Original-...	6619

Apartado B

Actualización e inserción de datos

```sql
DROP TABLE IF EXISTS part;
CREATE TABLE part
(
  P_PARTKEY BIGINT NOT NULL,
  P_NAME VARCHAR(55),
  P_MFGR VARCHAR(25),
  P_BRAND VARCHAR(10),
  P_TYPE VARCHAR(25),
  P_SIZE INTEGER,
  P_CONTAINER VARCHAR(10),
  P_RETAILPRICE DECIMAL(18,4),
  P_COMMENT VARCHAR(23)
)
diststyle ALL;
```

Result 1 | Result 2

Summary

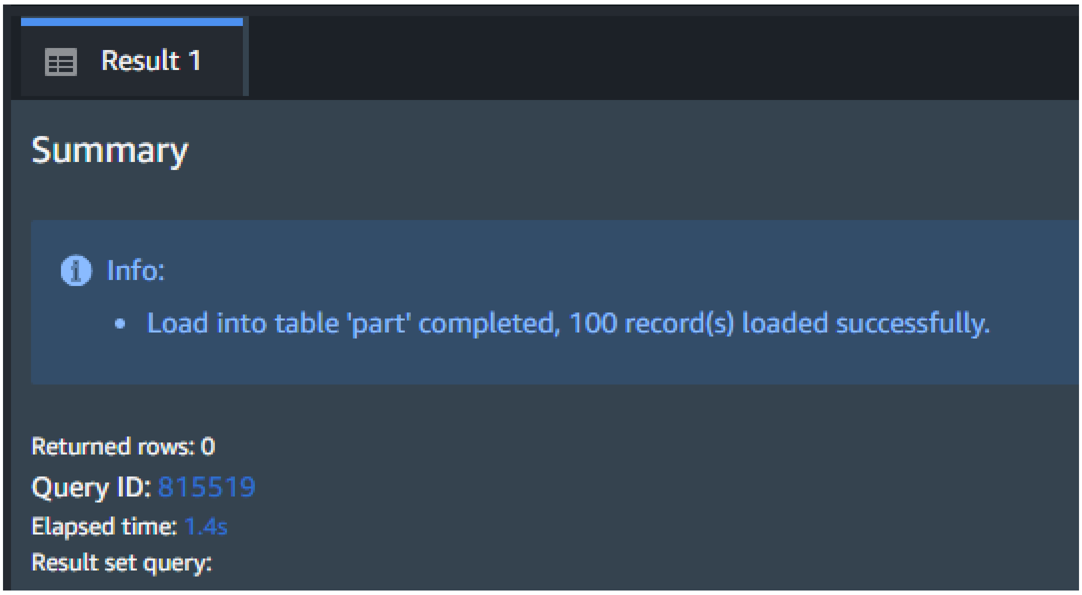
Returned rows: 0

Elapsed time: 389ms

Result set query:

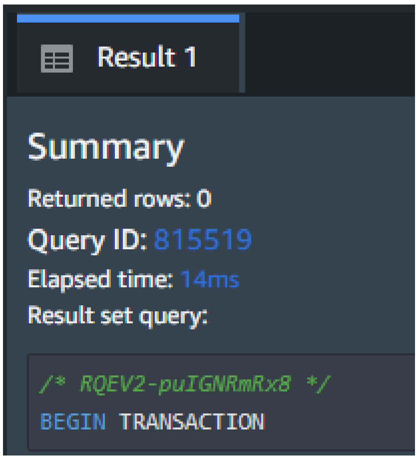
```sql
/* RQEV2-vfhBteXJHV */
CREATE TABLE part
(
 P_PARTKEY BIGINT NOT NULL,
 P_NAME VARCHAR(55),
 P_MFGR VARCHAR(25),
 P_BRAND VARCHAR(10),
 P_TYPE VARCHAR(25),
 P_SIZE INTEGER,
 P_CONTAINER VARCHAR(10),
 P_RETAILPRICE DECIMAL(18,4),
 P_COMMENT VARCHAR(23)
)
diststyle ALL
```

```sql
COPY part
FROM 's3://amazon-redshift-bucket-043356869404/Datos_Redshift/tbl/part.tbl'
iam_role 'arn:aws:iam::043356869404:role/LabRole'
DELIMITER '|'
IGNOREHEADER 0
REMOVEQUOTES
TRIMBLANKS
EMPTYASNULL
BLANKSASNULL
REGION 'us-east-1';
```



Ahora vamos a iniciar la transacción para actualizar la tabla de dimensiones “part”:

```sql
BEGIN TRANSACTION;
```



Creamos la tabla provisional:

```sql
DROP TABLE IF EXISTS stg_part;
CREATE TABLE stg_part
(
 NAME VARCHAR(55),
 MFGR VARCHAR(25),
 BRAND VARCHAR(10),
 TYPE VARCHAR(25),
 SIZE INTEGER,
 CONTAINER VARCHAR(10),
 RETAILPRICE DECIMAL(18,4),
 COMMENT VARCHAR(23)
)
BACKUP NO
;
COPY stg_part
FROM 's3://amazon-redshift-bucket-043356869404/Datos_Redshift/etl/part/' iam_role 'arn:aws:iam::043356869404:role/LabRole'csv gzip compupdate preset;
```

Result 1 | Result 2 | Result 3

Summary

Info:

- Load into table 'stg_part' completed, 10 record(s) loaded successfully.

Returned rows: 0

Query ID: 815573

Elapsed time: 1.3s

Result set query:

Y ahora vamos a actualizar los registros existentes:

```sql
-- 1️⃣ Actualizar registros existentes
UPDATE part
SET
  p_mfgr = stg_part.mfgr,
  p_brand = stg_part.brand,
  p_type = stg_part.type,
  p_size = stg_part.size,
  p_container = stg_part.container,
  p_retailprice = stg_part.retailprice,
  p_comment = stg_part.comment
FROM stg_part
WHERE part.p_name = stg_part.name;
```

Result 1 | Result 2

Summary

Returned rows: 0

Query ID: 815625

Elapsed time: 491ms

Result set query:

```sql
/* RQEV2-HdKFyoswz1 */
-- 1 Actualizar registros existentes
UPDATE part
SET
 p_mfgr = stg_part.mfgr,
 p_brand = stg_part.brand,
 p_type = stg_part.type,
 p_size = stg_part.size,
 p_container = stg_part.container,
 p_retailprice = stg_part.retailprice,
 p_comment = stg_part.comment
FROM stg_part
WHERE part.p_name = stg_part.name
```

```sql
-- 2 Insertar registros nuevos
INSERT INTO part (
  p_partkey, p_name, p_mfgr, p_brand, p_type, p_size,
  p_container, p_retailprice, p_comment
)
SELECT
  ROW_NUMBER() OVER () + COALESCE((SELECT MAX(p_partkey) FROM part), 0),
  stg_part.name,
  stg_part.mfgr,
  stg_part.brand,
  stg_part.type,
  stg_part.size,
  stg_part.container,
  stg_part.retailprice,
  stg_part.comment
FROM stg_part
LEFT JOIN part
  ON part.p_name = stg_part.name
WHERE part.p_name IS NULL;
```

Result 1 | Result 2

Summary

Affected rows: 10

Query ID: 815803

Elapsed time: 527ms

Result set query:

```sql
/* RQEV2-7kNyAtqj3r */
-- [2] Insertar registros nuevos
INSERT INTO part (
  p_partkey, p_name, p_mfgr, p_brand, p_type, p_size,
  p_container, p_retailprice, p_comment
)
SELECT
  ROW_NUMBER() OVER () + COALESCE((SELECT MAX(p_partkey) FROM part), 0),
  stg_part.name,
  stg_part.mfgr,
  stg_part.brand,
  stg_part.type,
  stg_part.size,
  stg_part.container,
  stg_part.retailprice,
  stg_part.comment
FROM stg_part
LEFT JOIN part
  ON part.p_name = stg_part.name
WHERE part.p_name IS NULL
```

Finalmente, vamos a terminar y confirmar la transacción:

```sql
1    END TRANSACTION;
```

Result 1

Summary

Returned rows: 0

Query ID: 815803

Elapsed time: 377ms

Result set query:

```sql
/* RQEV2-ODOAJQXpLD */
END TRANSACTION
```

Vamos a iniciar una nueva transacción:

```
BEGIN TRANSACTION;
```

Result 1

Summary

Returned rows: 0

Query ID: 815803

Elapsed time: 9ms

Result set query:

```
/* RQEV2-rpmJRI3Mux */
BEGIN TRANSACTION
```

```
-- Drop stg_lineitem if exists
DROP TABLE IF EXISTS stg_lineitem;
-- Create a stg_lineitem staging table and COPY data from input S3 location with the refreshed incremental data
CREATE TABLE stg_lineitem(
 orderkey BIGINT,
 LINENUMBER INTEGER NOT NULL,
 QUANTITY DECIMAL(18,4),
 EXTENDEDPRICE DECIMAL(18,4),
 DISCOUNT DECIMAL(18,4),
 TAX DECIMAL(18,4),
 RETURNFLAG VARCHAR(1),
 LINESTATUS VARCHAR(1),
 SHIPDATE DATE,
 COMMITDATE DATE,
 RECEIPTDATE DATE,
 SHIPINSTRUCT VARCHAR(25),
 SHIPMODE VARCHAR(10),
 COMMENT VARCHAR(44),
 p_name VARCHAR(55),
 s_name VARCHAR(25)
)
BACKUP NO;

COPY stg_lineitem
FROM 's3://amazon-redshift-bucket-043356869404/Datos_Redshift/etl/lineitem/' iam_role 'arn:aws:iam::043356869404:role/LabRole'csv gzip;
```

Result 1 | Result 2 | Result 3

Summary

Info:

- Load into table 'stg_lineitem' completed, 1000 record(s) loaded successfully.

Returned rows: 0

Query ID: 815858

Elapsed time: 1.1s

Result set query:

```sql
-- Se elimina cualquier fila de destino para la fecha de entrada para idempotencia
DELETE FROM lineitem WHERE l_shipdate = '2020-08-15';
```

Result 1

Summary

Returned rows: 0

Query ID: 815870

Elapsed time: 201ms

Result set query:

```sql
/* RQEV2-a3nfJGgAo7 */
-- Se elimina cualquier fila de destino para la fecha de entrada para idempotencia
DELETE FROM lineitem WHERE l_shipdate = '2020-08-15'
```

```sql
BEGIN;
DELETE FROM lineitem
USING (
 SELECT
 src.orderkey,
 p.p_partkey,
 s.s_suppkey
 FROM stg_lineitem src
 JOIN part p ON p.p_name = src.p_name
 JOIN supplier s ON s.s_name = src.s_name
 WHERE src.shipdate = '2020-08-15'
) d
```

```sql
WHERE lineitem.l_orderkey = d.orderkey
AND lineitem.l_partkey = d.p_partkey
AND lineitem.l_suppkey = d.s_suppkey;
INSERT INTO lineitem
SELECT
  src.orderkey,
  p.p_partkey,
  s.s_suppkey,
  src.linenumber,
  src.quantity,
  src.extendedprice,
  src.discount,
  src.tax,
  src.returnflag,
  src.linestatus,
  src.shipdate,
  src.commitdate,
  src.receiptdate,
  src.shipinstruct,
  src.shipmode,
  src.comment
FROM stg_lineitem src
JOIN part p ON p.p_name = src.p_name
JOIN supplier s ON s.s_name = src.s_name
WHERE src.shipdate = '2020-08-15';
END;
```

Result 1 | Result 2 | Result 3 | Result 4

Summary

Returned rows: 0

Query ID: 815944

Elapsed time: 369ms

Result set query:

Ahora confirmamos y finalizamos la transacción:

```sql
COMMIT;
```

Result 1

Summary

Returned rows: 0

Query ID: 815944

Elapsed time: 9ms

Result set query:

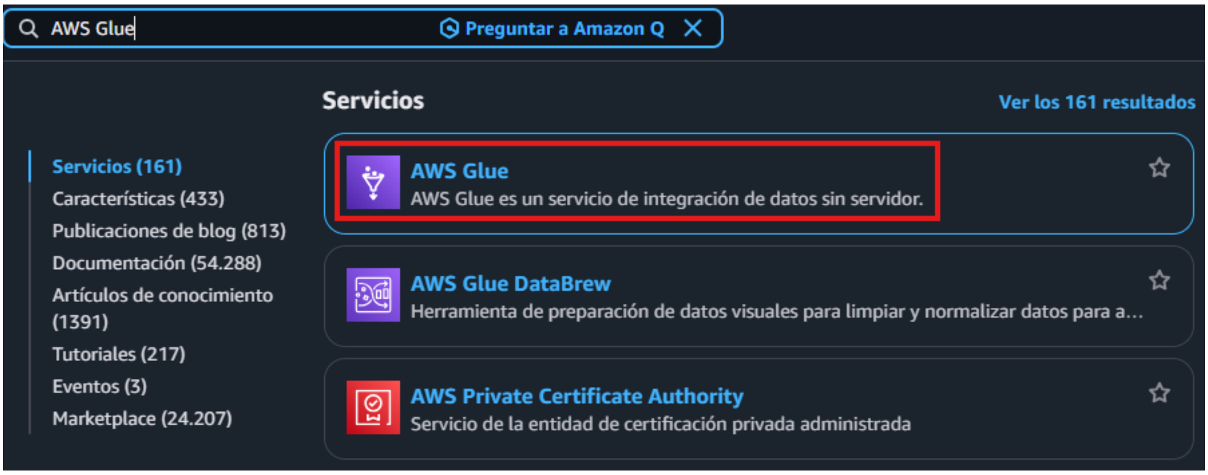
```sql
/* RQEV2-0eUiV2cRAg */
COMMIT
```

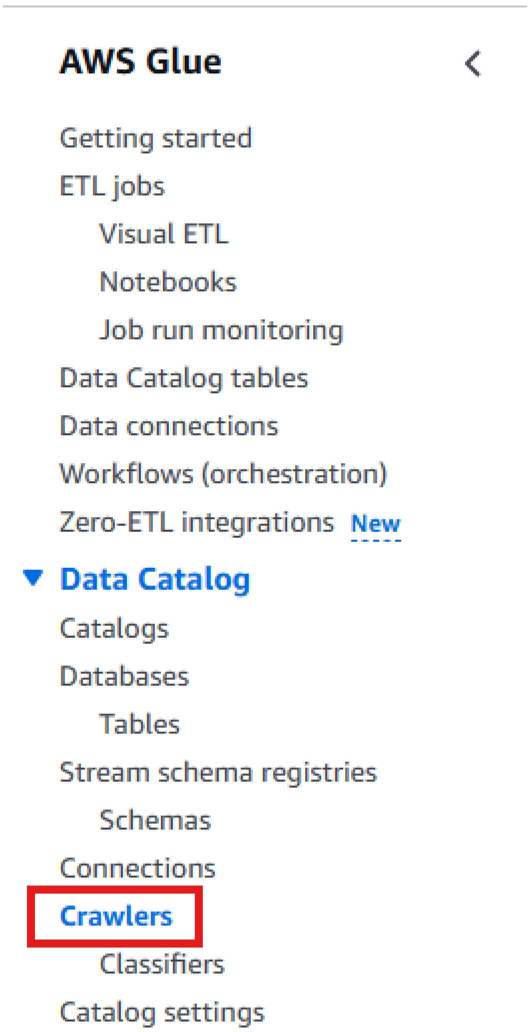
```sql
SELECT c_mktsegment,
 COUNT(o_orderkey) AS orders_count,
 SUM(l_quantity) AS quantity,
 SUM(l_extendedprice) AS extendedprice,
 COUNT(DISTINCT P_PARTKEY) AS parts_count,
 COUNT(DISTINCT L_SUPPKEY) AS supplier_count,
 COUNT(DISTINCT o_custkey) AS customer_count
FROM lineitem
 JOIN orders ON l_orderkey = o_orderkey
 JOIN customer c ON o_custkey = c_custkey
 JOIN part ON P_PARTKEY = l_PARTKEY
 JOIN supplier ON L_SUPPKEY = S_SUPPKEY
WHERE l_shipdate = '2020-08-15'
GROUP BY c_mktsegment;
```

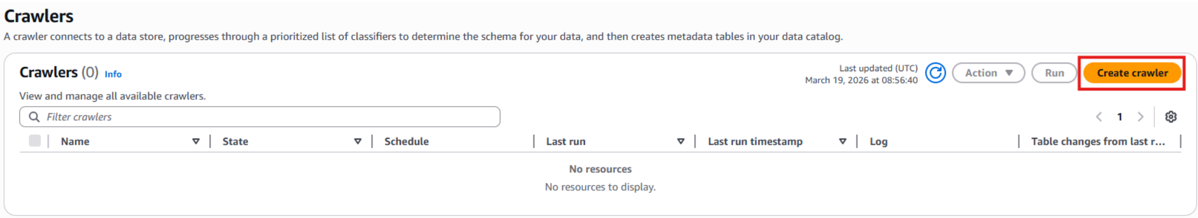
Apartado C

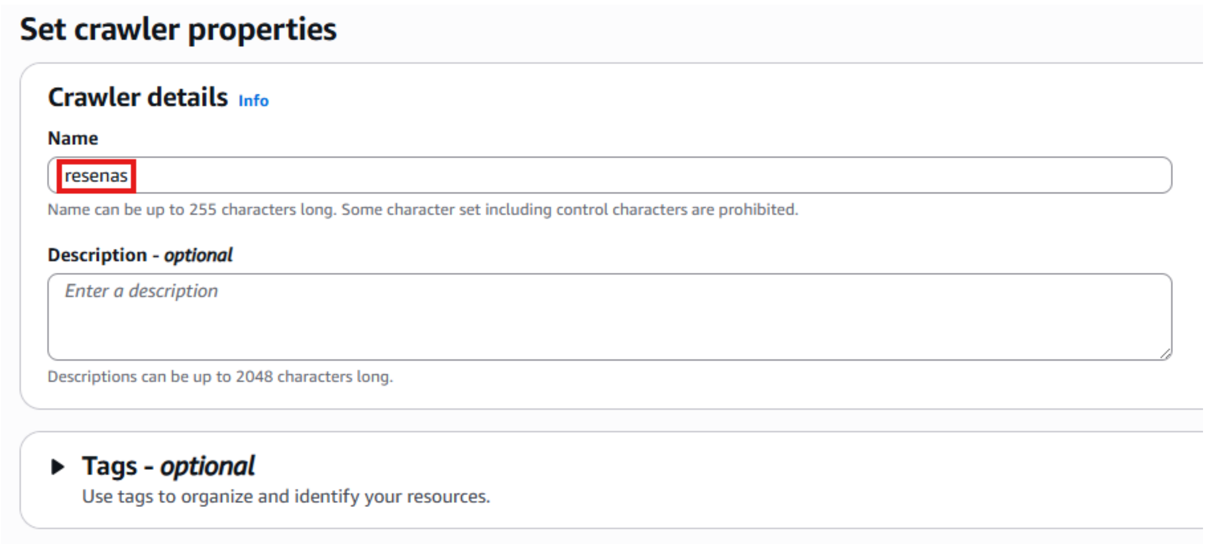
Catalogación e ingesta de datos con AWS Glue

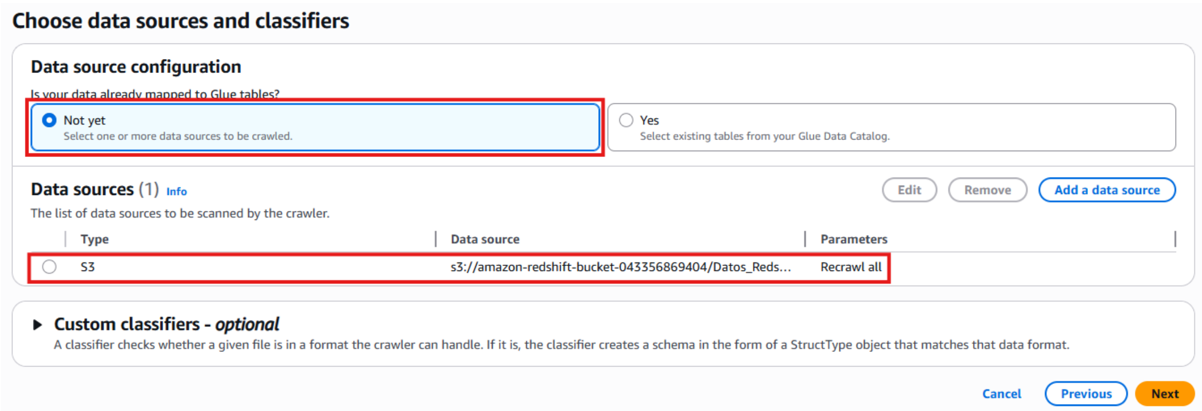
El primer paso es entrar en “AWS Glue”:

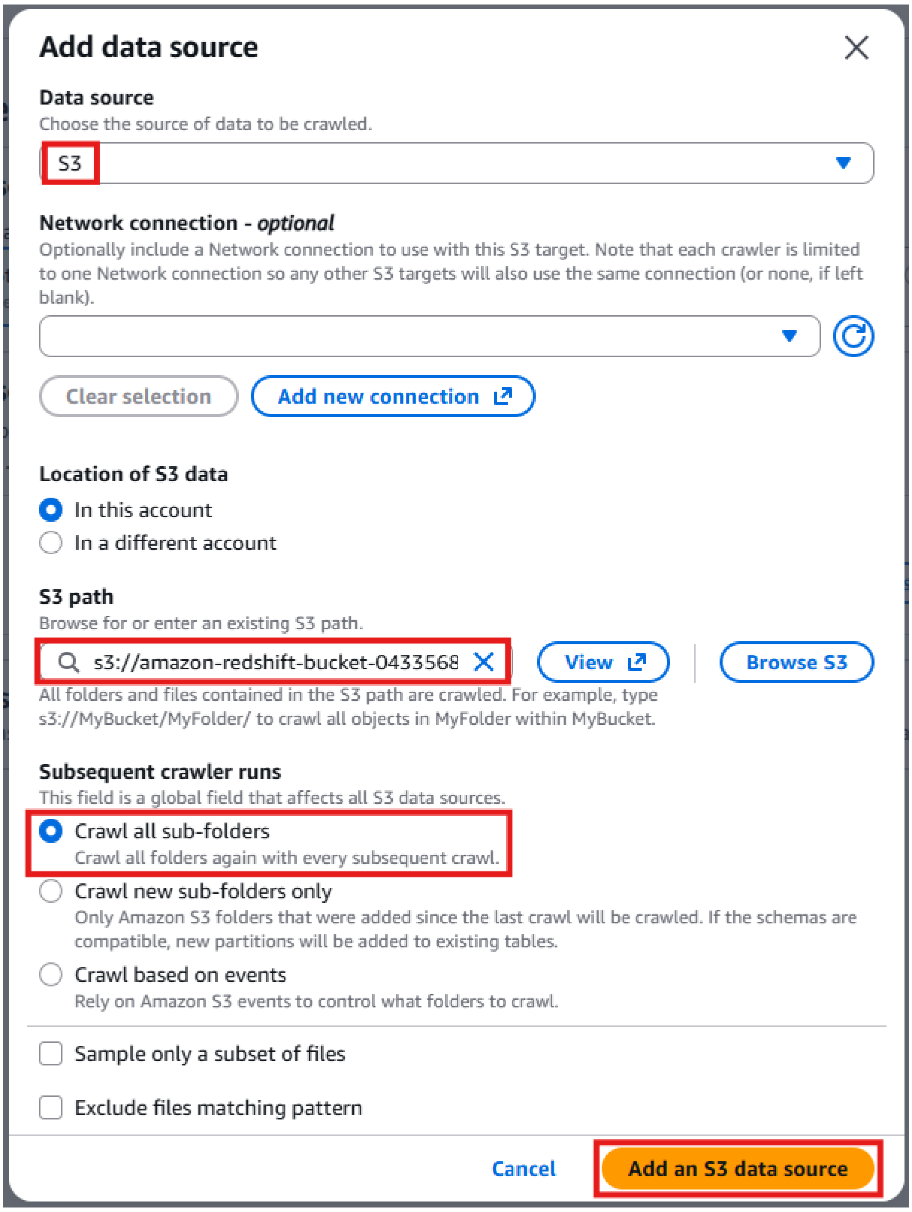


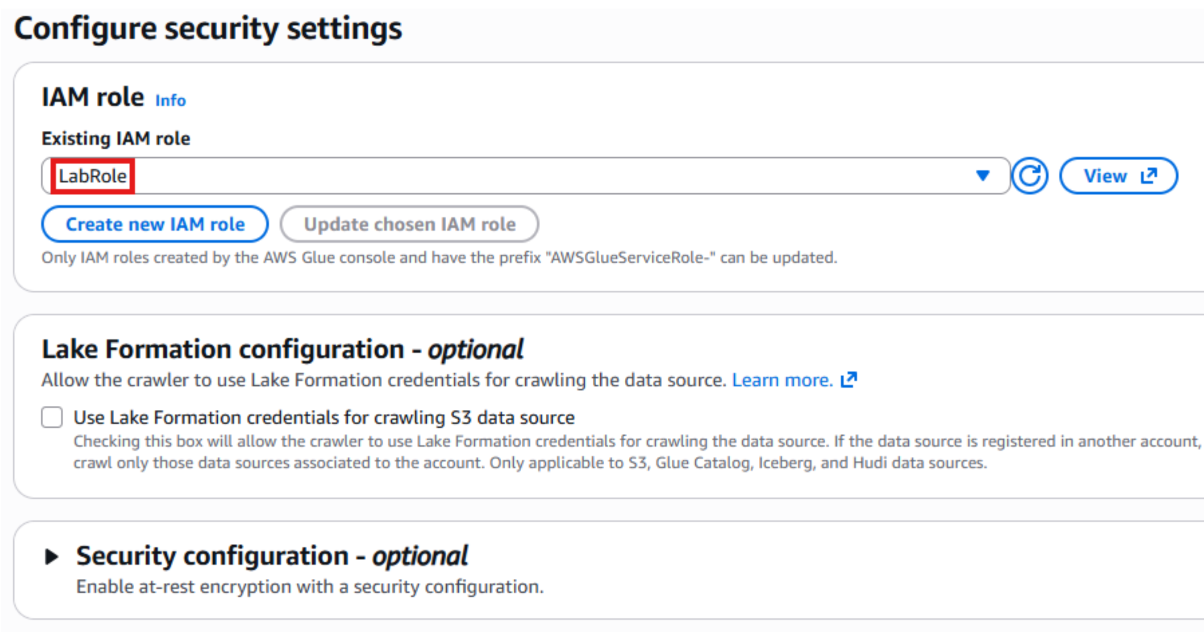



Crawlers
A crawler connects to a data store, progresses through a prioritized list of classifiers to determine the schema for your data, and then creates metadata tables in your data catalog.
Crawlers (0) Info
View and manage all available crawlers.
Last updated (UTC) March 19, 2026 at 08:56:40
Action
Run
Create crawler
Filter crawlers
Name
State
Schedule
Last run
Last run timestamp
Log
Table changes from last r...
No resources
No resources to display.


Set crawler properties
Crawler details Info
Name
resenas
Name can be up to 255 characters long. Some character set including control characters are prohibited.
Description - optional
Enter a description
Descriptions can be up to 2048 characters long.
Tags - optional
Use tags to organize and identify your resources.


Choose data sources and classifiers
Data source configuration
Is your data already mapped to Glue tables?
Not yet
Select one or more data sources to be crawled.
Yes
Select existing tables from your Glue Data Catalog.
Data sources (1) Info
The list of data sources to be scanned by the crawler.
Edit
Remove
Add a data source
Type
Data source
Parameters
S3
s3://amazon-redshift-bucket-043356869404/Datos_Reds...
Recrawl all
Custom classifiers - optional
A classifier checks whether a given file is in a format the crawler can handle. If it is, the classifier creates a schema in the form of a StructType object that matches that data format.
Cancel
Previous
Next


Add data source
Data source
Choose the source of data to be crawled.
S3
Network connection - optional
Optionally include a Network connection to use with this S3 target. Note that each crawler is limited to one Network connection so any other S3 targets will also use the same connection (or none, if left blank).
Clear selection
Add new connection
Location of S3 data
In this account
In a different account
S3 path
Browse for or enter an existing S3 path.
s3://amazon-redshift-bucket-0433568
View
Browse S3
All folders and files contained in the S3 path are crawled. For example, type s3://MyBucket/MyFolder/ to crawl all objects in MyFolder within MyBucket.
Subsequent crawler runs
This field is a global field that affects all S3 data sources.
Crawl all sub-folders
Crawl all folders again with every subsequent crawl.
Crawl new sub-folders only
Only Amazon S3 folders that were added since the last crawl will be crawled. If the schemas are compatible, new partitions will be added to existing tables.
Crawl based on events
Rely on Amazon S3 events to control what folders to crawl.
Sample only a subset of files
Exclude files matching pattern
Cancel
Add an S3 data source


Configure security settings
IAM role Info
Existing IAM role
LabRole
View
Create new IAM role
Update chosen IAM role
Only IAM roles created by the AWS Glue console and have the prefix "AWSGlueServiceRole-" can be updated.
Lake Formation configuration - optional
Allow the crawler to use Lake Formation credentials for crawling the data source. Learn more.
Use Lake Formation credentials for crawling S3 data source
Checking this box will allow the crawler to use Lake Formation credentials for crawling the data source. If the data source is registered in another account, crawl only those data sources associated to the account. Only applicable to S3, Glue Catalog, Iceberg, and Hudi data sources.
Security configuration - optional
Enable at-rest encryption with a security configuration.

Set output and scheduling

Output configuration Info

Target database

Choose a database

Clear selection | Add database

Table name prefix - *optional*

Type a prefix added to table names

Maximum table threshold - *optional*

This field sets the maximum number of tables the crawler is allowed to generate. In the event that this number is surpassed, the crawl will fail with an error. If depending on the data schema.

Type a number greater than 0

▶ **Advanced options**

Crawler schedule

You can define a time-based schedule for your crawlers and jobs in AWS Glue. The definition of these schedules uses the Unix-like cron

Frequency

On demand

Create a database

Create a database in the AWS Glue Data Catalog.

Database type

- (●) Glue Database
  Create a standard Glue database in the Data Catalog.
- ( ) Glue Database Resource Link
  Create a resource link to a database in another Data Catalog.
- ( ) Glue Database in S3 Tables Federated Catalog
  Create a database in an S3 Tables federated catalog.

Database details

Name

reviews

Database name is required, in lowercase characters, and no longer than 255 characters.

Description - *optional*

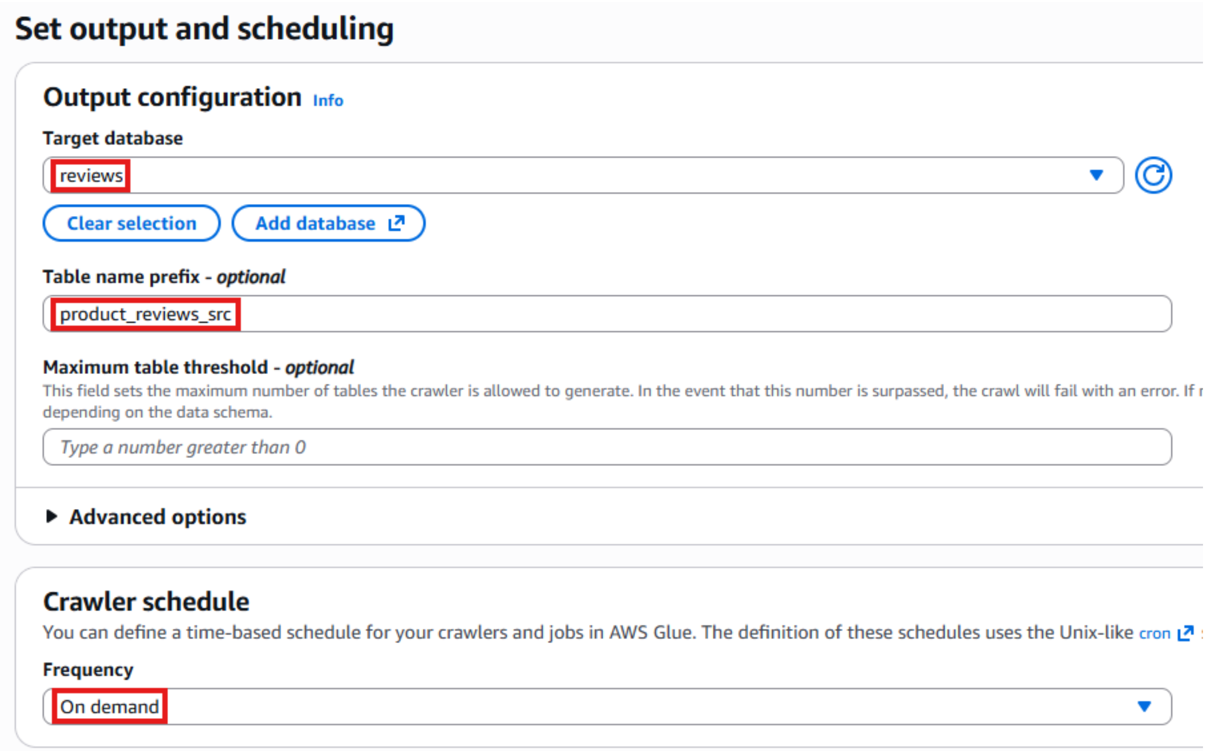
Enter text

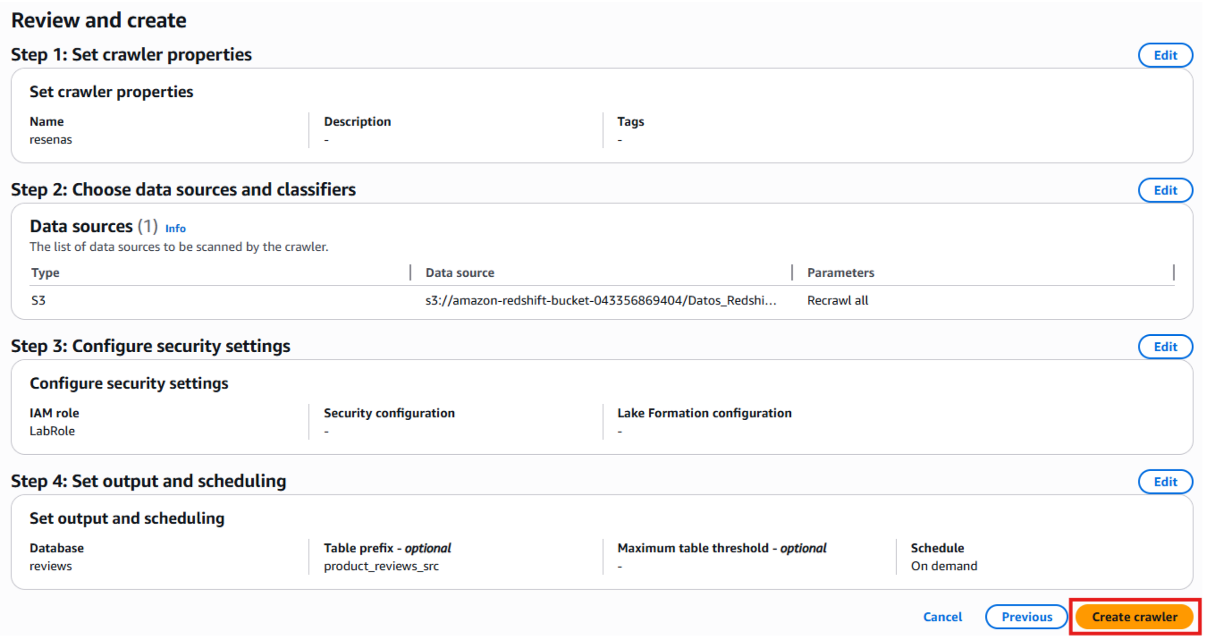
Descriptions can be up to 2048 characters long.

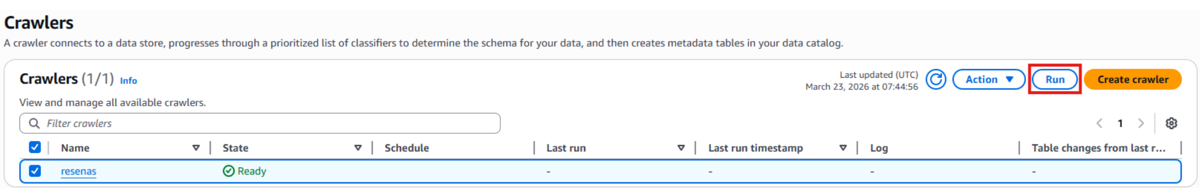
Location - *optional*

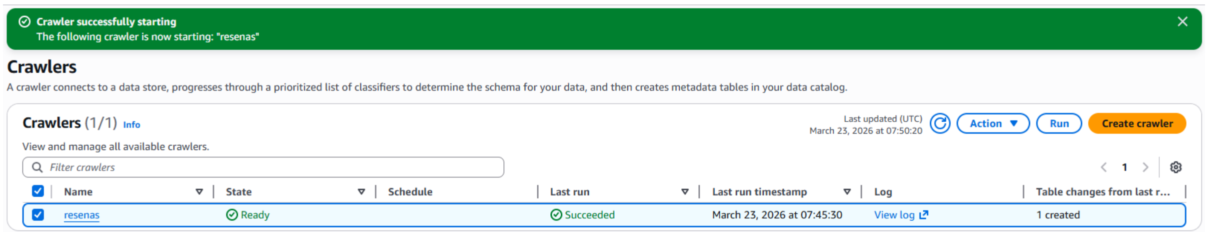
Set the URI location for use by clients of the Data Catalog.

An S3 location is required for managed tables and Zero-ETL integrations.

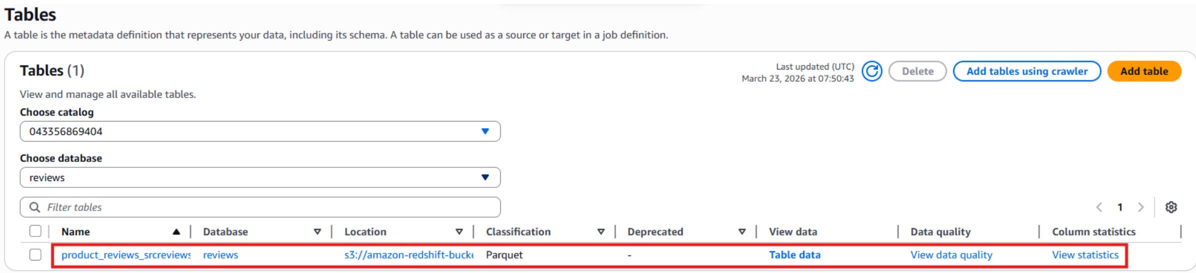

Set output and scheduling
Output configuration Info
Target database
reviews
Clear selection
Add database
Table name prefix - optional
product_reviews_src
Maximum table threshold - optional
This field sets the maximum number of tables the crawler is allowed to generate. In the event that this number is surpassed, the crawl will fail with an error. If r
depending on the data schema.
Type a number greater than 0
Advanced options
Crawler schedule
You can define a time-based schedule for your crawlers and jobs in AWS Glue. The definition of these schedules uses the Unix-like cron
Frequency
On demand


Review and create
Step 1: Set crawler properties
Edit
Set crawler properties
Name
resenas
Description
-
Tags
-
Step 2: Choose data sources and classifiers
Edit
Data sources (1) Info
The list of data sources to be scanned by the crawler.
Type
Data source
Parameters
S3
s3://amazon-redshift-bucket-043356869404/Datos_Redshi...
Recrawl all
Step 3: Configure security settings
Edit
Configure security settings
IAM role
LabRole
Security configuration
-
Lake Formation configuration
-
Step 4: Set output and scheduling
Edit
Set output and scheduling
Database
reviews
Table prefix - optional
product_reviews_src
Maximum table threshold - optional
-
Schedule
On demand
Cancel
Previous
Create crawler

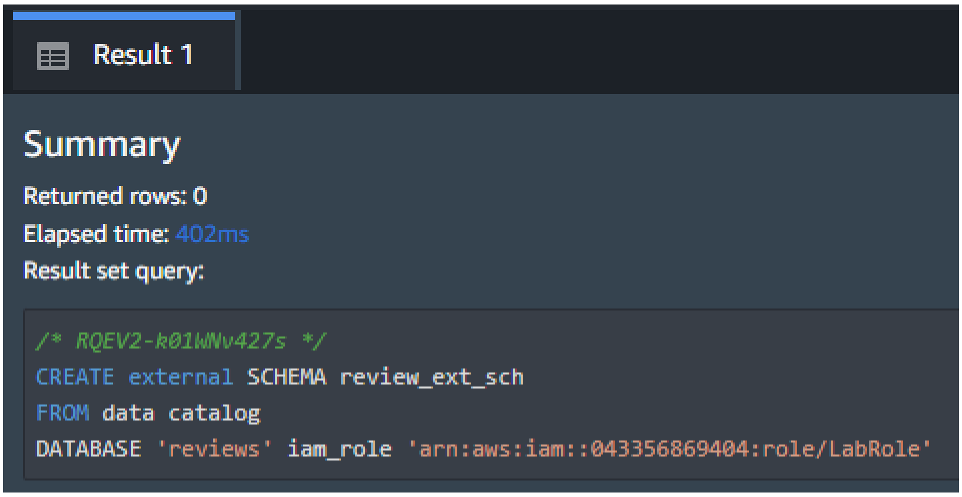

Crawlers
A crawler connects to a data store, progresses through a prioritized list of classifiers to determine the schema for your data, and then creates metadata tables in your data catalog.
Crawlers (1/1) Info
Last updated (UTC)
March 23, 2026 at 07:44:56
Action
Run
Create crawler
View and manage all available crawlers.
Filter crawlers
1
Name
State
Schedule
Last run
Last run timestamp
Log
Table changes from last r...
resenas
Ready
-
-
-
-

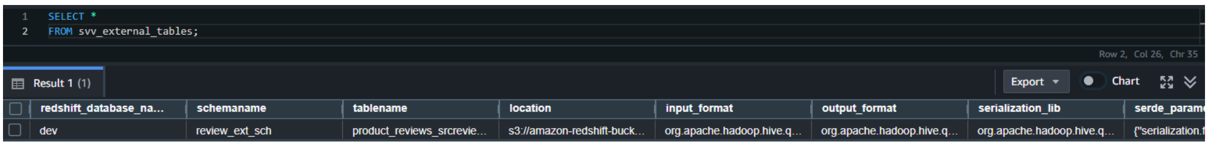


Ahora para verificar la tabla nos vamos a ir al apartado de “tablas”:




```sql
CREATE external SCHEMA review_ext_sch
FROM data catalog
DATABASE 'reviews' iam_role 'arn:aws:iam::043356869404:role/LabRole'
```





```sql
CREATE TABLE product_reviews_stage
(
 marketplace VARCHAR(2),
 customer_id VARCHAR(32),
 review_id VARCHAR(24),
 product_id VARCHAR(24),
 product_parent VARCHAR(32),
 product_title VARCHAR(512),
 star_rating INT,
 helpful_votes INT,
 total_votes INT,
 vine CHAR(1),
 verified_purchase CHAR(1),
 review_headline VARCHAR(256),
 review_body VARCHAR(MAX),
 review_date DATE,
 YEAR INT
)
DISTSTYLE KEY DISTKEY (customer_id) SORTKEY (review_date);
```

```sql
INSERT INTO product_reviews_stage
(
 marketplace,
 customer_id,
 review_id,
 product_id,
 product_parent,
 product_title,
 star_rating,
 helpful_votes,
 total_votes,
 vine,
 verified_purchase,
 review_headline,
 review_body,
 review_date,
 year
)
SELECT marketplace,
 customer_id,
 review_id,
 product_id,
 product_parent,
 product_title,
 star_rating,
 helpful_votes,
 total_votes,
 vine,
 verified_purchase,
 review_headline,
 review_body,
 review_date,
 year
FROM review_ext_sch.product_reviews_srcreviews_parquet;
```

```sql
SELECT * FROM product_reviews_stage;
```

Row 1, Col 37, Chr 36

Result 1 (100) Export Chart

marketplace	customer_id	review_id	product_id	product_parent	product_title	star_rating	helpful_vot
US	47651641	RIPDUN6PASB0H	B000050B0Q	263589486	Pirate Accessory Kit	5	0
US	46200620	R3KBPMZQPKZCU3	B00001OGXK	456099016	Disguise Austin Powers D...	4	4
US	51942478	R28LV9KBQU52DX	B00005OT7L	90669056	Women's Black Button-Fr...	4	0
US	50031514	R3FRV569SBYKVL	B00005OT7L	90669056	Women's Black Button-Fr...	4	0
US	40253381	R2S71EFZSOFGCA	B00005OU5B	299588169	Men's Black Button-Front ...	5	0
US	51130970	R2B3RCLF2BT5YZ	B00005OU5B	299588169	Men's Black Button-Front ...	4	0
US	38845246	R1BNGSKYTMV65T	B00005OU5B	299588169	Men's Black Button-Front ...	4	0
US	51365767	RK1IR71PF2Z7P	B00005OU5B	299588169	Men's Black Button-Front ...	4	0
US	49818499	R2K24K6FM17WXC	B00005OT7L	90669056	Women's Black Button-Fr...	5	0
US	51361307	R2TH8P61EY5D3Q	B00005OT7L	90669056	Women's Black Button-Fr...	5	0
US	50898608	R32G5TO62W43KP	B00005OT7L	90669056	Women's Black Button-Fr...	5	0
US	50059103	R1UF0XDQBHPZM7	B00005OT7L	90669056	Women's Black Button-Fr...	5	0
US	38343758	R170PMRKGSA04J	B00005OT7L	90669056	Women's Black Button-Fr...	5	0
US	47177850	R3T1E8FWPJ20XJ	B00005OT7L	90669056	Women's Black Button-Fr...	5	0
US	45313777	R58KJMDZOXU7M	B00005OT7L	90669056	Women's Black Button-Fr...	5	0
US	48795807	R306CL8IDJ10R5	B00005OT7L	90669056	Women's Black Button-Fr...	5	0
US	38150730	R1EGOT7LQAF6DF	B00005OT7L	90669056	Women's Black Button-Fr...	4	0
US	37367215	RF2GW5ITRA3BP	B00005OT7L	90669056	Women's Black Button-Fr...	5	0
US	37095642	RI4RC7L66HATK	B00005OT7L	90669056	Women's Black Button-Fr...	5	0
US	51686706	R3K7A71ZE1TJXU	B000050696	282078585	Fun World Men 'Dripping ...	3	20
US	51404743	R19OZBA4I2FA6D	B00005OU5B	299588169	Men's Black Button-Front ...	5	0
US	38439544	R2Q41K33IRQ9F8	B00005OUH9	478722567	Women's Fleece Bucket ...	4	0
US	49094261	R1FSBCO167F24E	B00006A94I	183755563	Girls' Cardigan Sweater - ...	5	0
US	36898241	R4BV6ZHWS86F5	B00006FXJ6	422132618	Biblioholism Shirts	5	0

Apartado D

Creación directa de base de datos en Data Catalog desde Redshift

```sql
CREATE EXTERNAL SCHEMA spectrum_schema
FROM DATA CATALOG
DATABASE 'spectrum_db'
IAM_ROLE 'arn:aws:iam::043356869404:role/LabRole'
CREATE EXTERNAL DATABASE IF NOT EXISTS;
```

Result 1

Summary

Info:

- External database "spectrum_db" created

Returned rows: 0
Query ID: 1018710
Elapsed time: 472ms
Result set query:

Databases (2)

Last updated (UTC) March 23, 2026 at 08:14:49 — Edit — Delete — Add database

A database is a set of associated table definitions, organized into a logical group.

Choose catalog

043356869404

Filter databases

Name	Description	Location URI	Source catalog	Created on (UTC)
reviews	-	-	043356869404	March 19, 2026 at 09:16:05
spectrum_db	-	-	043356869404	March 23, 2026 at 08:14:00

```sql
CREATE EXTERNAL TABLE spectrum_schema.reviews_parquet
(
 marketplace VARCHAR(2),
 customer_id VARCHAR(32),
 review_id VARCHAR(24),
 product_id VARCHAR(24),
 product_parent VARCHAR(32),
 product_title VARCHAR(512),
 star_rating INT,
 helpful_votes INT,
 total_votes INT,
 vine CHAR(1),
 verified_purchase CHAR(1),
 review_headline VARCHAR(256),
 review_body VARCHAR(MAX),
 review_date DATE,
 YEAR INT
)
STORED AS PARQUET
LOCATION 's3://amazon-redshift-bucket-043356869404/Datos_Redshift/reviews_parquet/';
```

Tables

A table is the metadata definition that represents your data, including its schema. A table can be used as a source or target in a job definition.

Tables (1)

Last updated (UTC)
March 23, 2026 at 08:19:19

Delete | Add tables using crawler | Add table

View and manage all available tables.

Choose catalog

043356869404

Choose database

spectrum_db

Filter tables

< 1 >

Name	Database	Location	Classification	Deprecated	View data	Data quality	Column statistics
reviews_parquet	spectrum_db	s3://amazon-redshift-buck(	-	-	Table data	View data quality	View statistics

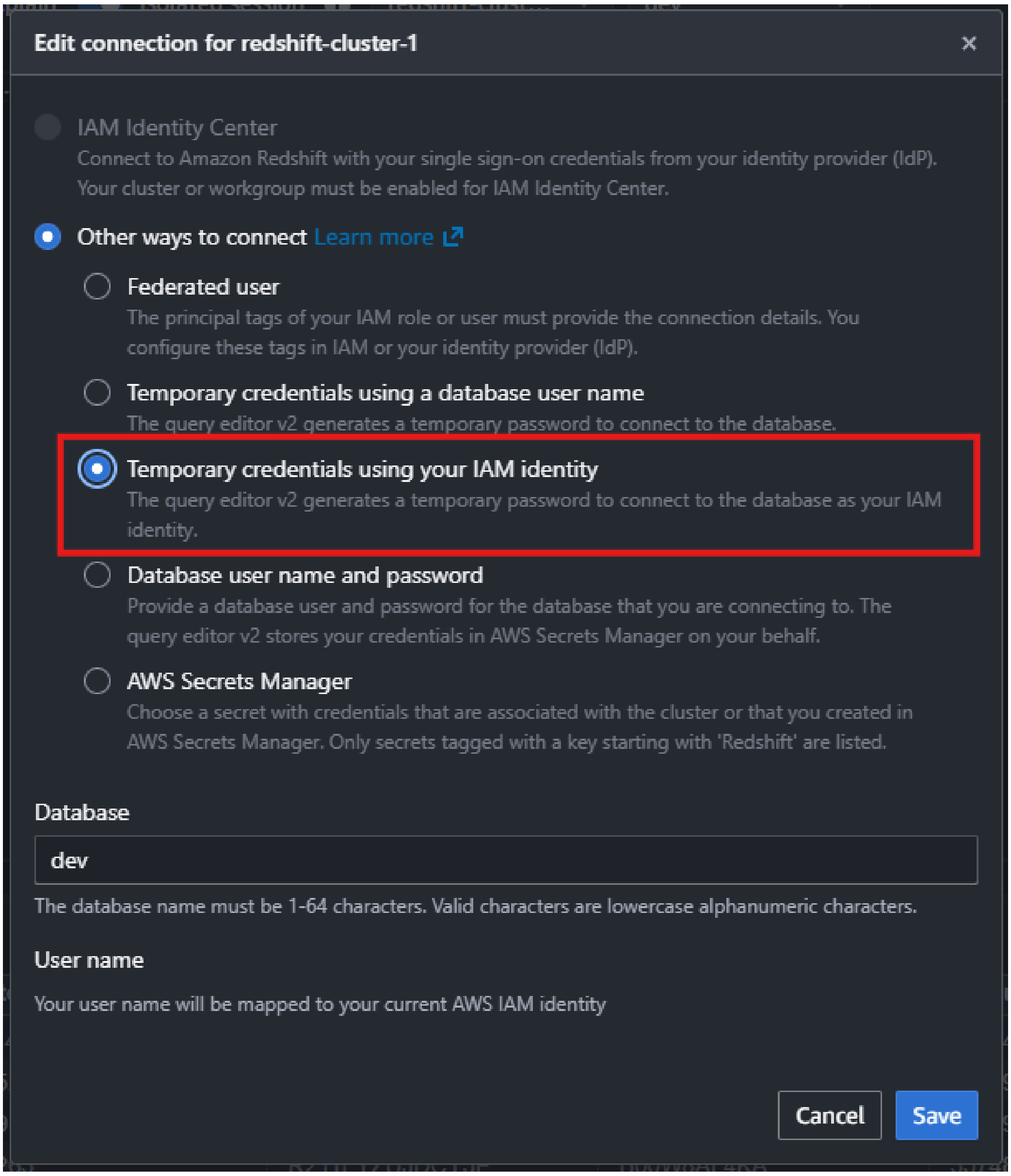
```sql
SELECT *
FROM "dev"."spectrum_schema"."reviews_parquet";
```

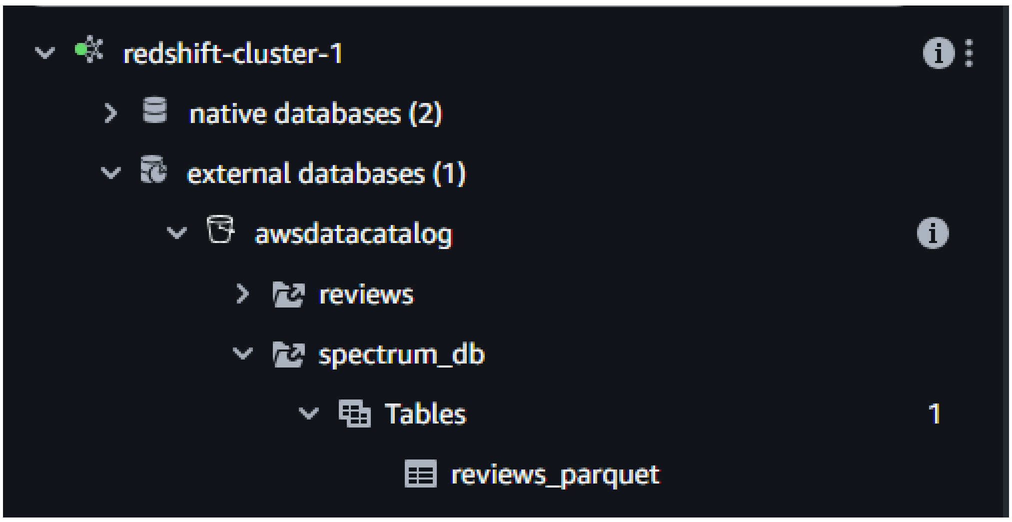
Row 2, Col 6, Chr 57

Result 1 (100)

Export | Chart

marketplace	customer_id	review_id	product_id	product_parent	product_title	star_rating	helpful_
US	51114360	R2O1ZWACRT6PGH	B00004U3M4	988649596	Adult Robin Costume Del...	1	9
US	17052567	R26I9SLC2PZTW7	B00WGDJJM4	973297679	LookbookStore Women's ...	5	2
US	50898608	R32G5TO62W43KP	B00005OT7L	90669056	Women's Black Button-Fr...	5	0
US	627263	R211FYZUJDCY3P	B00W8AL4KA	357485050	Nintendo Zelda triforce lo...	5	0
US	35515711	RBQ22581QAV0U	B0000644X6	101524150	Marc New York Leather E...	4	0
US	53017806	R2B5IGV10Y92E3	B00VWGXVTI	57069564	Polare Full Grain Leather ...	5	9
US	35360489	R26B0PZGWNYB3S	B00006B6YD	757094727	Dogs Playing Poker Micro...	5	0
US	451719	RL6X6IE6D6PRG	B00VQMQ69U	633742649	USA Flag T-Shirt Skull Me...	5	0
US	35360489	R2TXWGT5KI15OL	B00006FXDZ	512328322	Mind, Body, Soul Outfit	4	0
US	25693810	R1LYBT7I02B8V3	B00VMU5JVM	159861482	Sassy Apparel Womens E...	5	0
US	36772555	R4K29O35NKCN9	B00006FXOE	809038758	Sports Sweaters	5	0
US	11144467	R26DCOB4JDOS0P	B00VLMXP8K	934268644	Sexy Womens 2 Pieces ...	4	2
US	34449191	R3K0D1GL20O1AA	B00006JKXO	632656242	Women's Mossimo Mens...	5	0
US	37989735	R14NZO9P3DIWIM	B00VLAS5IC	908196049	Sexy Women Strappy Ca...	1	10
US	52547797	R1Q6KOPJLUN039	B000068YC0	807307180	Edward Gorey "There's N...	5	0
US	14964432	R17W3IFEWLLCH4	B00VKAUP7W	801479285	Pink Queen® Women'...	1	1
US	52958624	R2D1NESDBFSNNY	B000079XY0	836742467	Santa Maria Novella Polle...	4	0
US	25982580	R3KG6WN19KLE1Q	B00VHCGAL8	2431327	TWINTH Dolman Drape T...	3	0
US	52791587	R3BCWGLCYNZ64E	B00007266Y	579633929	Lands' End Women's Reg...	5	7
US	13134123	R3VKD4LDGEGSHT	B00VGM12N0	573696169	Kathlena Neoprene Divin...	2	0
US	40666092	R14BUCDYUVH6UR	B00006WZQT	237353922	OshKosh Purple/Pink Sno...	5	0
US	11266069	R248GQDDTVTNQC	B00VG4P8H4	304822894	Boys Disney Cars Lightni...	2	0


Edit connection for redshift-cluster-1
IAM Identity Center
Connect to Amazon Redshift with your single sign-on credentials from your identity provider (IdP). Your cluster or workgroup must be enabled for IAM Identity Center.
Other ways to connect Learn more
Federated user
The principal tags of your IAM role or user must provide the connection details. You configure these tags in IAM or your identity provider (IdP).
Temporary credentials using a database user name
The query editor v2 generates a temporary password to connect to the database.
Temporary credentials using your IAM identity
The query editor v2 generates a temporary password to connect to the database as your IAM identity.
Database user name and password
Provide a database user and password for the database that you are connecting to. The query editor v2 stores your credentials in AWS Secrets Manager on your behalf.
AWS Secrets Manager
Choose a secret with credentials that are associated with the cluster or that you created in AWS Secrets Manager. Only secrets tagged with a key starting with 'Redshift' are listed.
Database
dev
The database name must be 1-64 characters. Valid characters are lowercase alphanumeric characters.
User name
Your user name will be mapped to your current AWS IAM identity
Cancel
Save

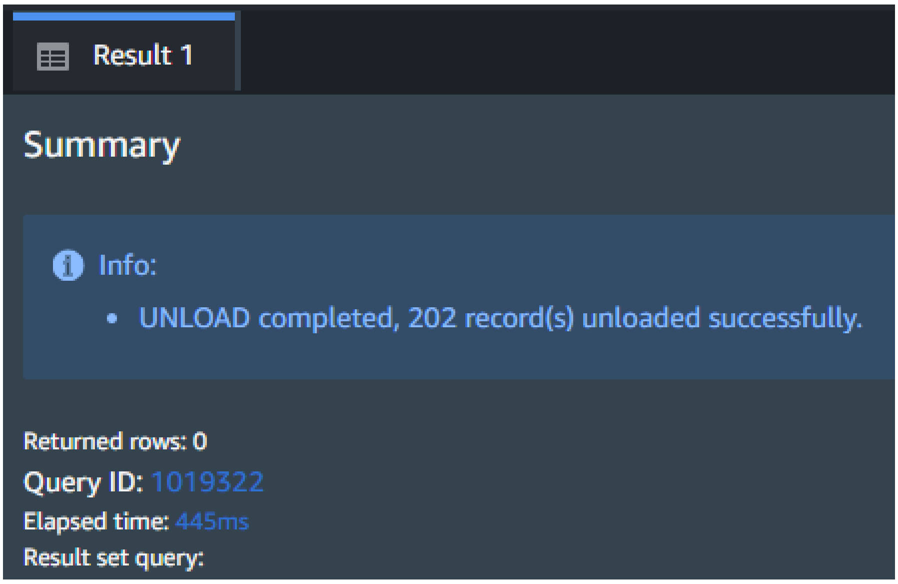

redshift-cluster-1
native databases (2)
external databases (1)
awsdatacatalog
reviews
spectrum_db
Tables
1
reviews_parquet

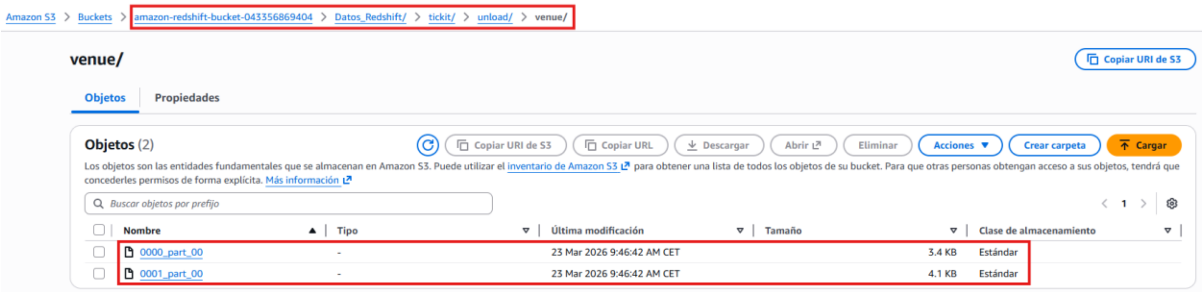
Apartado E

Exportar datos de AWS Redshift a AWS S3

Ahora vamos a enviar el contenido de la tabla VENUE al bucket de Amazon S3:

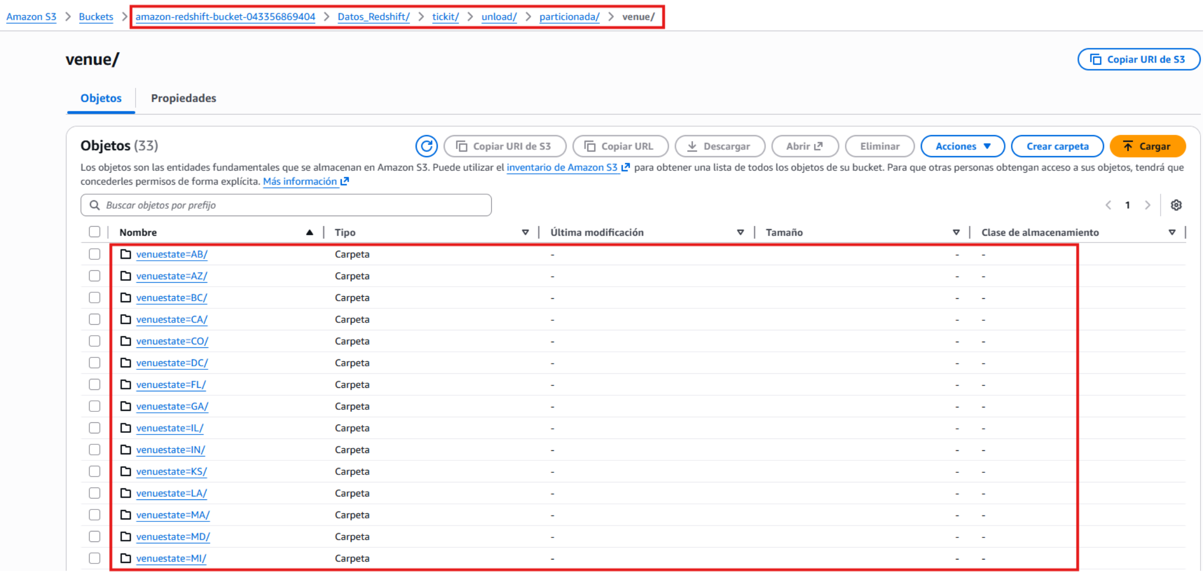
```
unload ('select * from tickit.venue')
to 's3://amazon-redshift-bucket-043356869404/Datos_Redshift/tickit/unload/venue/'
iam_role 'arn:aws:iam::043356869404:role/LabRole'
ALLOWOVERWRITE
```





Ahora vamos a exportarla en formato CSV particionada:

```
unload ('select * from tickit.venue')
to 's3://amazon-redshift-bucket-043356869404/Datos_Redshift/tickit/unload/particionada/venue/'
iam_role 'arn:aws:iam::043356869404:role/LabRole'
CSV
PARTITION BY (venuestate)
ALLOWOVERWRITE
```


Amazon S3 > Buckets > amazon-redshift-bucket-043356869404 > Datos_Redshift/ > tickit/ > unload/ > particionada/ > venue/
venue/
Copiar URI de S3
Objetos
Propiedades
Objetos (33)
Copiar URI de S3
Copiar URL
Descargar
Abrir
Eliminar
Acciones
Crear carpeta
Cargar
Los objetos son las entidades fundamentales que se almacenan en Amazon S3. Puede utilizar el inventario de Amazon S3 para obtener una lista de todos los objetos de su bucket. Para que otras personas obtengan acceso a sus objetos, tendrá que concederles permisos de forma explícita. Más información
Buscar objetos por prefijo
Nombre
Tipo
Última modificación
Tamaño
Clase de almacenamiento
venuestate=AB/
venuestate=AZ/
venuestate=BC/
venuestate=CA/
venuestate=CO/
venuestate=DC/
venuestate=FL/
venuestate=GA/
venuestate=IL/
venuestate=IN/
venuestate=KS/
venuestate=LA/
venuestate=MA/
venuestate=MD/
venuestate=MI/
Carpeta